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(54) **LARGE LANGUAGE MODEL-PREDICTED RESPONSE BIASING FOR CONVERSATIONAL SYSTEMS**

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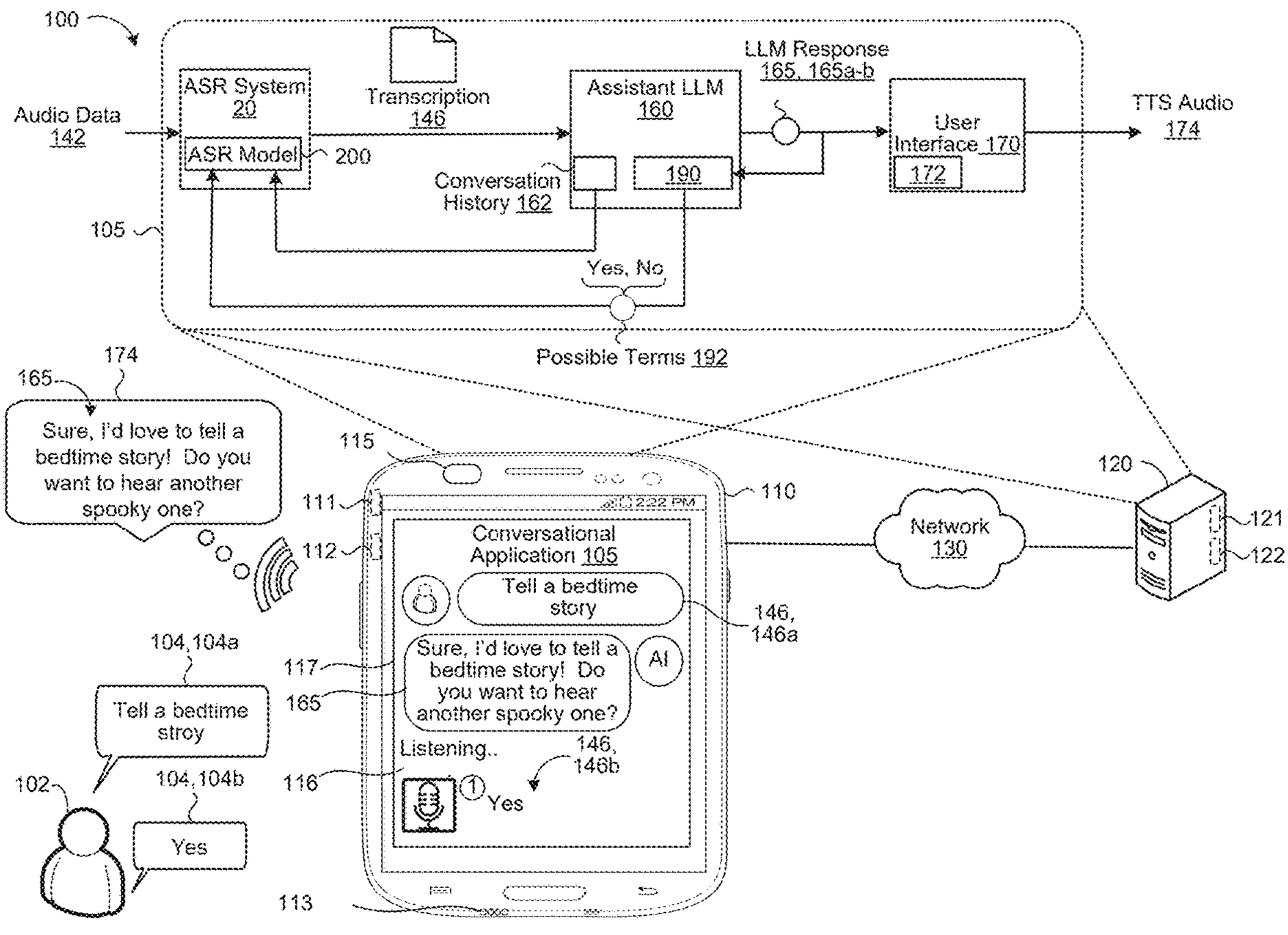
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(57) **ABSTRACT**

A method includes receiving a response generated by an assistant LLM that is directed toward the user during an assistant turn in conversation between a user and the assistant LLM. Based on the response generated by the assistant LLM, the method also includes predicting, by the assistant LLM, one or more possible terms the user may speak in a follow-up query to the response generated by the assistant LLM. The method also includes biasing an automated speech recognition (ASR) model toward recognizing the one or more possible terms predicted by the assistant LLM, receiving audio data characterizing the follow-up query to the response generated by the assistant LLM during the user turn subsequent to the assistant turn in the voice-based conversation, and processing, using the ASR model biased toward recognizing the one or more possible terms, the audio data to generate a transcription of the follow-up query.



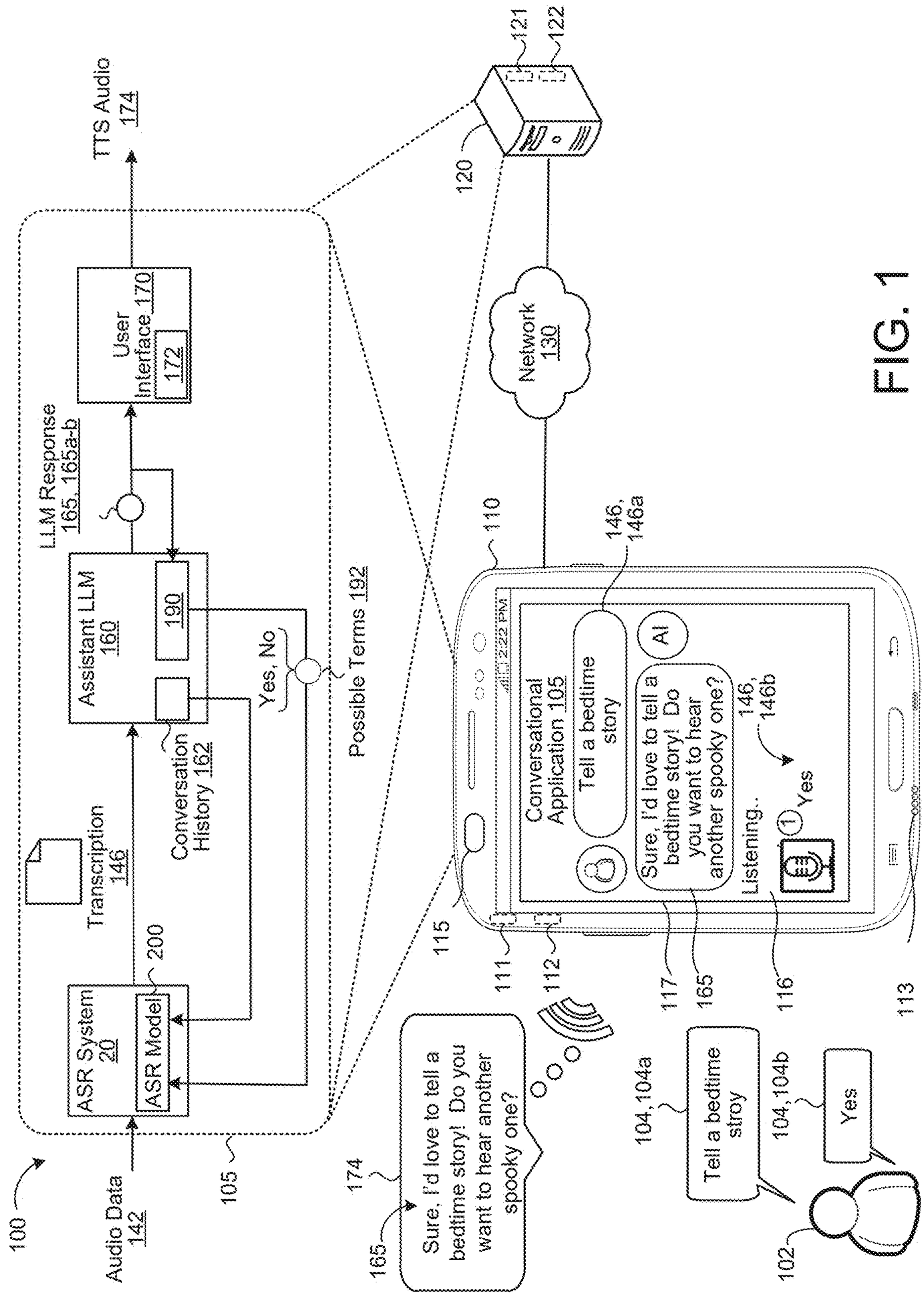

100
Audio Data 142
ASR System 20
ASR Model 206
105
Transcription 146
200
Assistant LLM 160
Conversation History 162
190
LLM Response 165, 165a-b
User Interface 170
172
TTS Audio 174
Yes, No
Possible Terms 192
120
121
122
Network 130
110
115
111
112
117
165
116
113
146, 146a
146, 146b
Conversational Application 105
Tell a bedtime story
Sure, I'd love to tell a bedtime story! Do you want to hear another spooky one?
AI
Listening..
Yes
2:22 PM
165
Sure, I'd love to tell a bedtime story! Do you want to hear another spooky one?
104,104a
Tell a bedtime stroy
104,104b
Yes
102

FIG. 1

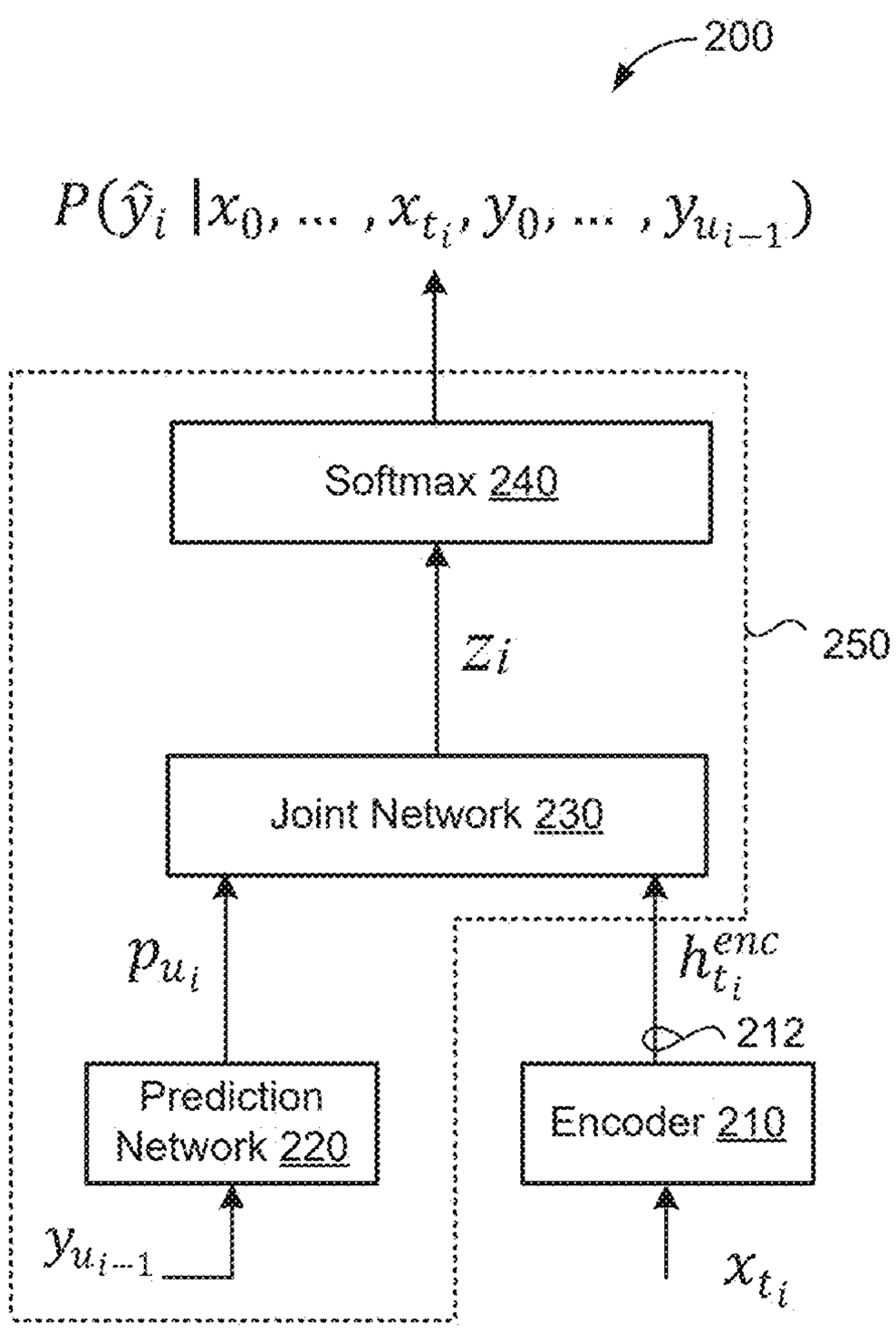

200
$P(\hat{y}_i | x_0, \cdots, x_{t_i}, y_0, \cdots, y_{u_{i-1}})$
Softmax 240
z_i
250
Joint Network 230
p_{u_i}
$h_{t_i}^{enc}$
212
Prediction Network 220
Encoder 210
$y_{u_{i-1}}$
x_{t_i}

FIG. 2

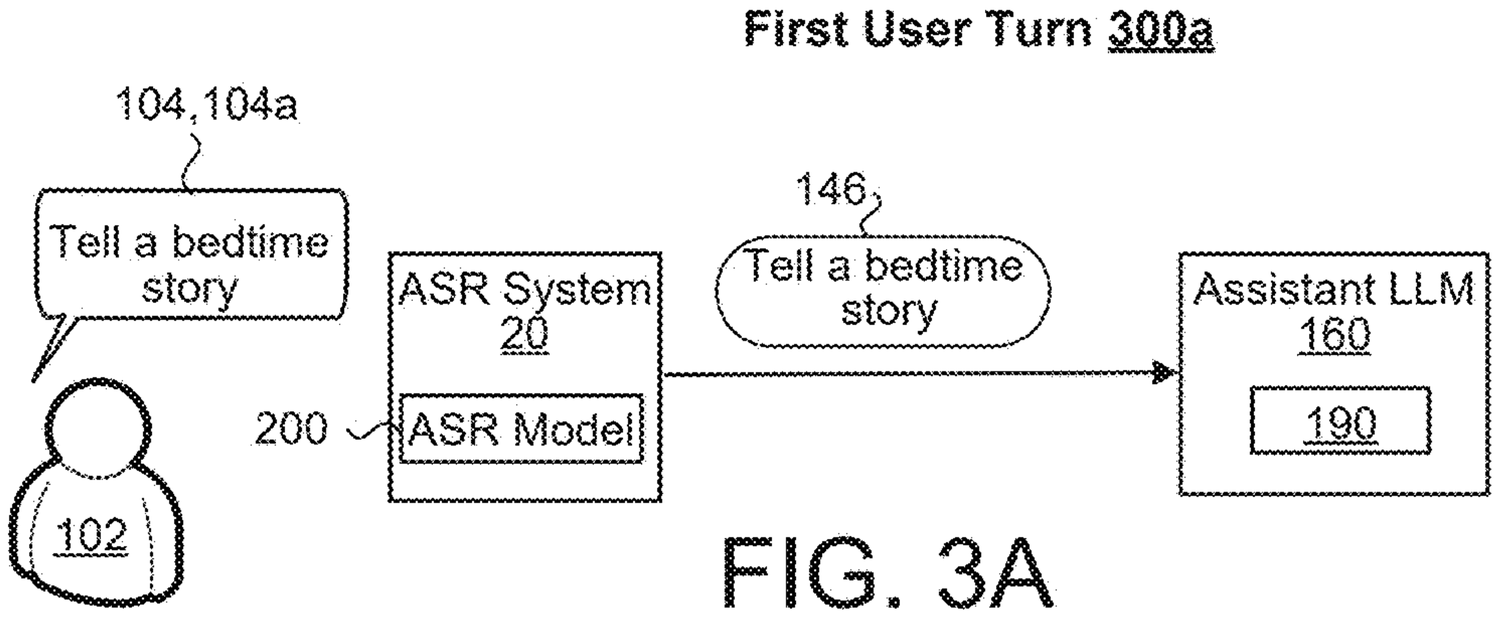


FIG. 3A

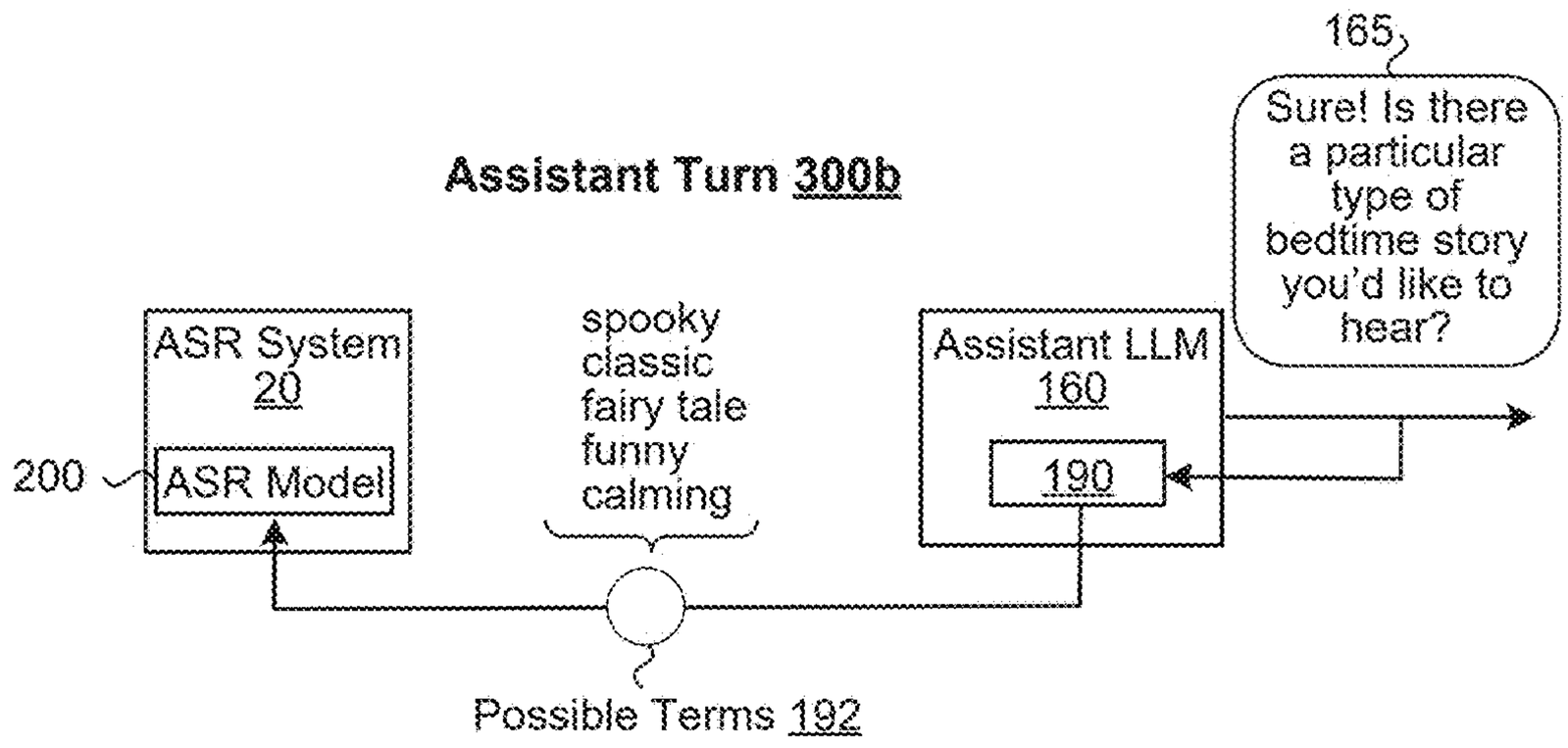


FIG. 3B

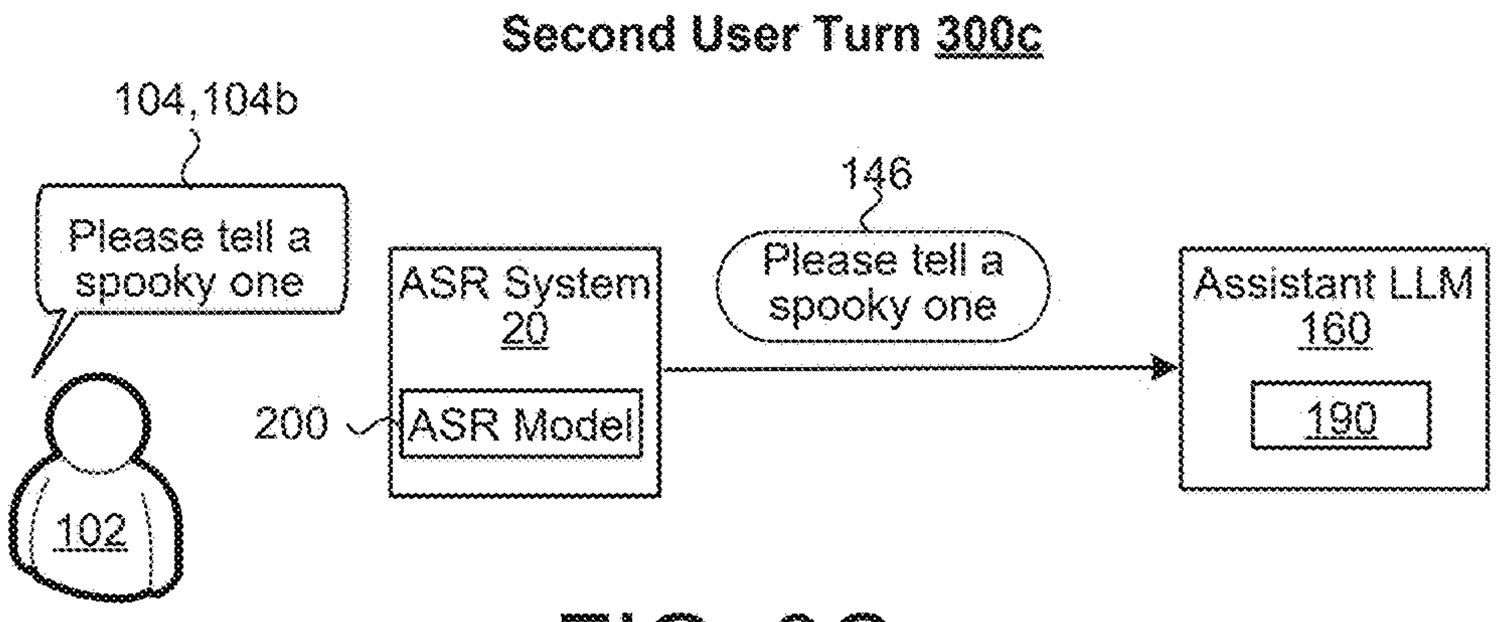


FIG. 3C

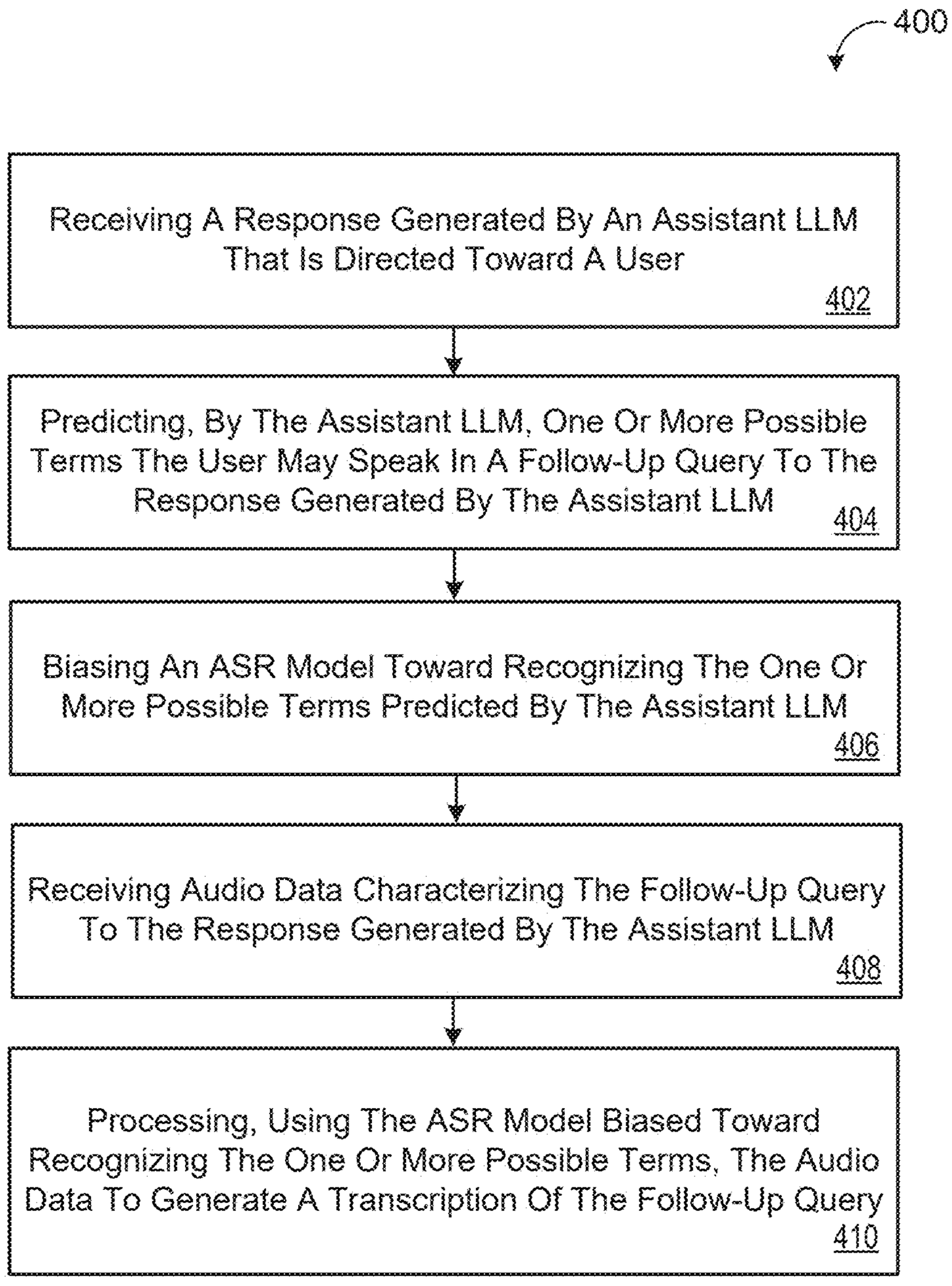

400
Receiving A Response Generated By An Assistant LLM That Is Directed Toward A User
402
Predicting, By The Assistant LLM, One Or More Possible Terms The User May Speak In A Follow-Up Query To The Response Generated By The Assistant LLM
404
Biasing An ASR Model Toward Recognizing The One Or More Possible Terms Predicted By The Assistant LLM
406
Receiving Audio Data Characterizing The Follow-Up Query To The Response Generated By The Assistant LLM
408
Processing, Using The ASR Model Biased Toward Recognizing The One Or More Possible Terms, The Audio Data To Generate A Transcription Of The Follow-Up Query
410

FIG. 4

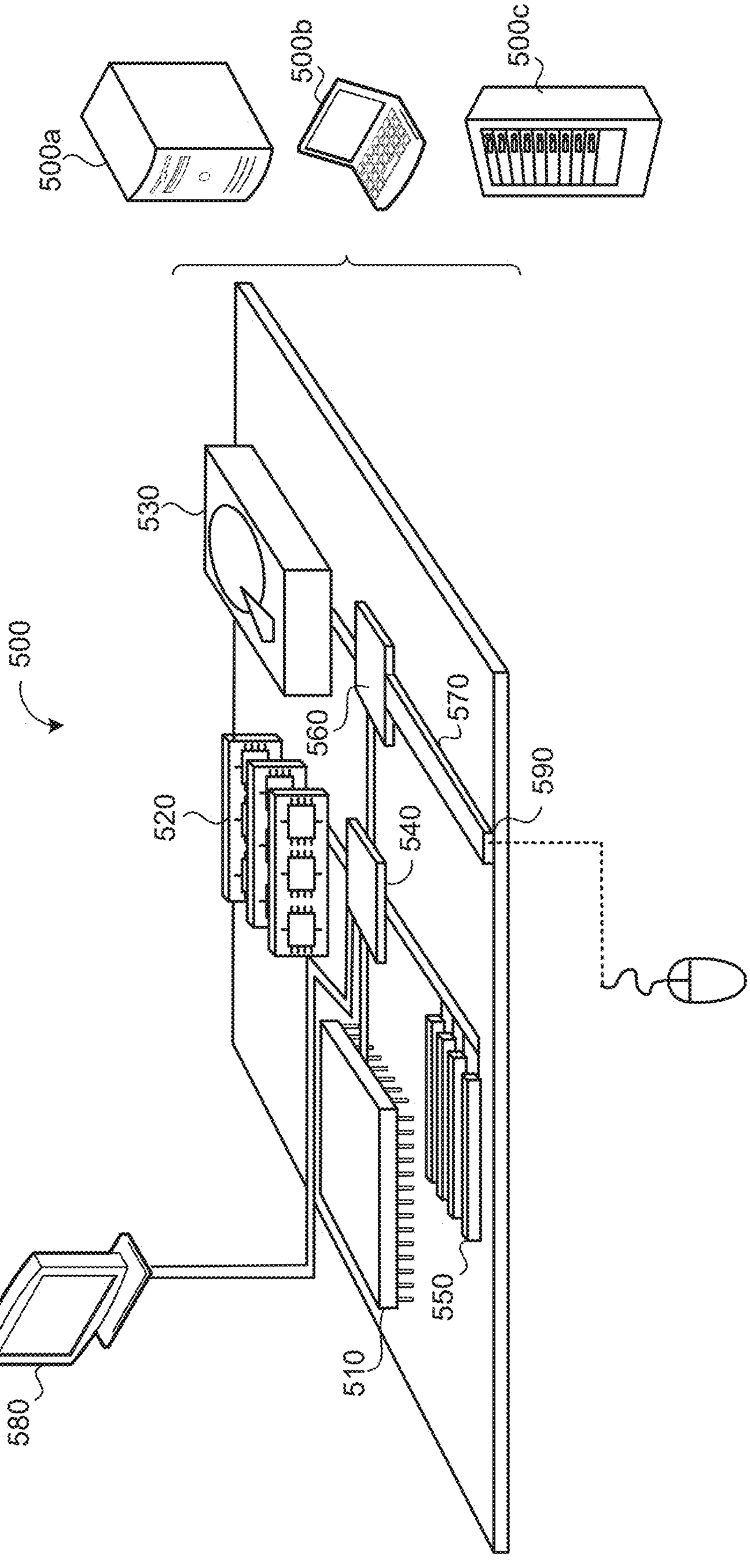


FIG. 5

LARGE LANGUAGE MODEL-PREDICTED RESPONSE BIASING FOR CONVERSATIONAL SYSTEMS

TECHNICAL FIELD

[0001] U.S. Patent application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application 63/752,407, filed on Jan. 31, 2025. The disclosure of this prior application is considered part of the disclosure of this application and is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates to large language model-predicted response biasing for conversational systems.

BACKGROUND

[0003] Conversational systems leveraging large language models (LLMs) as chatbots/assistants can allow voice-based conversations where a user can speak a query directed toward the LLM, listen to an audio-based response to the query returned from the LLM, and continue the voice-based conversation with the LLM. Often, these conversational systems include a cascaded architecture where an automated speech recognition (ASR) model is responsible for transcribing user speech into text, and then the ASR model sends the text to the LLM for processing to generate a response to the query. The cascaded architecture may further include a text-to-speech (TTS) system that converts a textual representation of the response generated by the LLM into TTS audio that conveys the response as synthesized speech for playback to the user.

[0004] In these cascaded architectures where a separate ASR model is responsible for providing the transcription of the user speech to the LLM, the ASR model lacks context from earlier in the conversation to produce the most accurate transcription. While the LLM itself may have a very clear prediction of what form the user's speech may take, the ASR model that will be transcribing the user's speech is agnostic to this information.

DESCRIPTION OF DRAWINGS

[0005] FIG. **1** is a schematic view of a system for biasing a speech recognition model when transcribing user speech in a voice-based conversation between the user and an assistant large language model (LLM).

[0006] recognition during a voice-based conversation between a user and a large language model (LLM)-powered assistant.

[0007] FIG. **2** is a schematic view of an example speech recognition model having a recurrent neural network-transducer architecture.

[0008] FIGS. **3**A-**3**C are schematic views of user and assistant turns during a voice-based conversation between a user and an assistant LLM.

[0009] FIG. **4** includes a flowchart of an example arrangement of operations for a computer-implemented method of biasing an ASR model during a voice-based conversation between the user and the assistant LLM.

[0010] FIG. **5** is a schematic view of an example computing device that may be used to implement the systems and methods described herein.

[0011] Like reference symbols in the various drawings indicate like elements.

SUMMARY

[0012] One aspect of the present disclosure provides a computer-implemented method executing on data processing hardware that causes the data processing hardware to perform operations that include, during an assistant turn in a voice-based conversation between a user and an assistant large language model (LLM), receiving a response generated by the assistant LLM that is directed toward the user, and based on the response generated by the assistant LLM that is directed toward the user, predicting, by the assistant LLM, one or more possible terms the user may speak in a follow-up query to the response generated by the assistant LLM during a user turn subsequent to the assistant turn in the voice-based conversation. The operations also include biasing an automated speech recognition (ASR) model toward recognizing the one or more possible terms predicted by the assistant LLM; during the user turn subsequent to the assistant turn in the voice-based conversation, receiving audio data characterizing the follow-up query to the response generated by the assistant LLM; and processing, using the ASR model biased toward recognizing the one or more possible terms predicted by the assistant LLM, the audio data to generate a transcription of the follow-up query.

[0013] Implementations of the disclosure may include one or more of the following optional features. In some implementations, predicting the one or more possible terms the user may speak in the follow-up query includes processing a textual representation of the response to: determine that the response directed toward the user comprises a question that solicits the user to speak an answer to the question; and identify a list of likely answers to the question, and determining the one or more possible terms the user may speak in the follow-up query as the list of likely answers to the question. In other implementations, predicting the one or more possible terms the user may speak in the follow-up query includes: processing a textual representation of the response to determine the response includes a question directed toward the user that solicits the user to speak a first term or a second term in the follow-up query; and determining the one or more possible terms the user may speak in the follow-up query comprises the first term and the second term. In these other implementations, the first term may include one of Yes or No, and the second term may include the other one of Yes or No. Further, the response including the question directed toward the user may solicit the user to speak the first term or the second term without specifying the first term and the second term in the response.

[0014] In some examples, the operations also include obtaining, from the assistant LLM, a conversation history of the voice-based conversation comprising all previous queries input by the user and corresponding responses returned by the assistant LLM during the voice-based conversation. In these examples, biasing the ASR model toward recognizing the one or more possible terms predicted by the assistant LLM further includes biasing the ASR model toward recognizing biasing terms related to the conversation history of the voice-based conversation. Here, the one or more possible terms predicted by the assistant LLM may be different from the biasing terms related to the conversation history.

[0015] In some implementations, the operations also include, after generating the transcription of the follow-up query, processing, by the assistant LLM, the transcription of the follow-up query to generate another response directed

toward the user that is responsive to the follow-up query. Additionally or alternatively, the operations may also include processing, by a text-to-speech (TTS) system, a textual representation of the response generated by assistant LLM to generate TTS audio characterizing a synthesized speech representation of the response, and providing, for audible output from a user device associated with the user, the TTS audio characterizing the synthesized speech representation of the response. The ASR model may include an acoustic encoder and a speech decoder, while the assistant LLM may include a plurality of pre-trained multi-head attention layers. The ASR model and the plurality of pre-trained multi-head attention layers may be trained separately.

[0016] Another aspect of the present disclosure includes a system having data processing hardware and memory hardware in communication with the data processing and storing instructions that when executed on the data processing hardware causes the data processing hardware to perform operations. The operations include, during an assistant turn in a voice-based conversation between a user and an assistant large language model (LLM), receiving a response generated by the assistant LLM that is directed toward the user, and based on the response generated by the assistant LLM that is directed toward the user, predicting, by the assistant LLM, one or more possible terms the user may speak in a follow-up query to the response generated by the assistant LLM during a user turn subsequent to the assistant turn in the voice-based conversation. The operations also include biasing an automated speech recognition (ASR) model toward recognizing the one or more possible terms predicted by the assistant LLM; during the user turn subsequent to the assistant turn in the voice-based conversation, receiving audio data characterizing the follow-up query to the response generated by the assistant LLM; and processing, using the ASR model biased toward recognizing the one or more possible terms predicted by the assistant LLM, the audio data to generate a transcription of the follow-up query.

[0017] This aspect may include one or more of the following optional features. In some implementations, predicting the one or more possible terms the user may speak in the follow-up query includes processing a textual representation of the response to: determine that the response directed toward the user comprises a question that solicits the user to speak an answer to the question; and identify a list of likely answers to the question, and determining the one or more possible terms the user may speak in the follow-up query as the list of likely answers to the question. In other implementations, predicting the one or more possible terms the user may speak in the follow-up query includes: processing a textual representation of the response to determine the response includes a question directed toward the user that solicits the user to speak a first term or a second term in the follow-up query; and determining the one or more possible terms the user may speak in the follow-up query comprises the first term and the second term. In these other implementations, the first term may include one of Yes or No, and the second term may include the other one of Yes or No. Further, the response including the question directed toward the user may solicit the user to speak the first term or the second term without specifying the first term and the second term in the response.

[0018] In some examples, the operations also include obtaining, from the assistant LLM, a conversation history of the voice-based conversation comprising all previous queries input by the user and corresponding responses returned by the assistant LLM during the voice-based conversation. In these examples, biasing the ASR model toward recognizing the one or more possible terms predicted by the assistant LLM further includes biasing the ASR model toward recognizing biasing terms related to the conversation history of the voice-based conversation. Here, the one or more possible terms predicted by the assistant LLM may be different from the biasing terms related to the conversation history.

[0019] In some implementations, the operations also include, after generating the transcription of the follow-up query, processing, by the assistant LLM, the transcription of the follow-up query to generate another response directed toward the user that is responsive to the follow-up query. Additionally or alternatively, the operations may also include processing, by a text-to-speech (TTS) system, a textual representation of the response generated by assistant LLM to generate TTS audio characterizing a synthesized speech representation of the response, and providing, for audible output from a user device associated with the user, the TTS audio characterizing the synthesized speech representation of the response. The ASR model may include an acoustic encoder and a speech decoder, while the assistant LLM may include a plurality of pre-trained multi-head attention layers. The ASR model and the plurality of pre-trained multi-head attention layers may be trained separately.

[0020] The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.

DETAILED DESCRIPTION

[0021] Humans may engage in human-to-computer dialogs with interactive software applications referred to as "chatbots," "voice bots", "automated assistants", "interactive personal assistants," "intelligent personal assistants," "conversational agents," etc. via a variety of computing devices. As one example, these chatbots may correspond to a machine learning model or a combination of different machine learning models, and may be utilized to perform various tasks on behalf of users.

[0022] Chatbots adopting Large language models (LLMs) are currently opening up a wide range of applications due to their powerful understanding and generation capabilities which can operate over text, image, and/or audio inputs. These models are also being extended with actuation capabilities via integration mechanisms with various service providers.

[0023] Automatic speech recognition (ASR) systems focus on providing not only high quality (e.g., a low word error rate), but also low latency (e.g., a short delay between a user speaking and a transcription or response appearing) speech recognition for spoken utterances. For example, when using a device that implements an ASR system, there is often an expectation that the ASR system decodes utterances in a streaming fashion that corresponds to real-time or even faster than real-time.

[0024] Conversational applications (e.g., digital assistants, chatbots, voice-controlled applications, etc.) leveraging large language models (LLMs) have become more popular

in recent years. Recently, the use of LLMs in conversational applications have been adapted to accept speech from a user that solicits a response from the LLM, and the conversational application in turn can provide the response generated by the LLM for audible output from a user device. These conversational applications adapted for voice-based conversations may leverage a cascaded architecture that includes an ASR model for transcribing user speech into text and the LLM for processing the text transcribed by the ASR model to generate a response. The cascaded architecture may further include a text-to-speech (TTS) system that converts a textual representation of the response generated by the LLM into TTS audio that conveys the response as synthesized speech. For instance, the user device can use a microphone to capture a natural language utterance spoken by the user, the ASR model can convert the spoken natural language utterance into text, the LLM extracts the user's intent from the text and generates a textual response based on the user's intent, and the TTS system converts the textual response into synthesized speech for audible output from the user device. In some scenarios, the ASR model includes an audio encoder that first encodes audio data characterizing the spoken input and a decoder that decodes the encoded audio data into text to form the transcription.

[0025] The LLM may maintain a conversation history of the voice-based conversation between the user that conveys the transcriptions of the natural language utterances spoken by the user and the corresponding textual responses generated by the LLM. The conversational application may use the conversation history to bias the ASR model toward recognizing biasing terms conveyed earlier in the conversation history to contextualize the transcription of user speech during a current user turn in the voice-based conversation. However, after the LLM generates a response during an assistant turn in the voice-based conversation, the LLM may be able to ascertain a very clear prediction of what form the user's speech may take in a follow-up query to the response that is not available to the ASR model that will be transcribing the user's speech.

[0026] Implementations herein are directed toward leveraging the LLM to predict possible terms a user may speak during a next user turn in a voice-based conversation between the user and an assistant LLM based on a response generated by the LLM during an assistant turn in the voice-based conversation and using the possible terms predicted by the LLM to bias the ASR model toward recognizing the possible terms in speech spoken by the user during the next user turn. Specifically, the conversational application may receive a response generated by the assistant LLM that is directed toward the user during an assistant turn in the voice-based conversation, and based on the response, the LLM may predict one or more possible terms the user may speak in a follow-up query to the response generated by the assistant LLM during a user turn subsequent to the assistant turn in the voice-based conversation. Thereafter, the conversational application may bias the ASR model toward recognizing the one or more possible terms predicted by the assistant so that when the conversational application receives audio data characterizing the follow-up query during the user turn subsequent to the assistant turn in the voice-based conversational application, the ASR model biased toward recognizing the one or more possible terms predicted by the assistant LLM may process the audio data to generate an accurate transcription of the follow-up query.

[0027] Notably, while the ASR model may optionally be further biased toward recognizing biasing terms related to a conversation history of the voice-based conversation maintained by the assistant LLM, these biasing terms relate to entities previously recited during the conversation history, and are thus, different than the one or more possible terms that the LLM predicts the user is likely to speak in the follow-query based on the response generated by the LLM. For instance, the LLM may determine that the response includes a question directed toward the user that solicits the user to speak "Yes" or "No" as an answer to the question. Here, the LLM may predict the terms "Yes" and "No" as possible terms the user is likely to speak in a follow-up query for use in biasing the ASR model toward recognizing the terms "Yes" and "No" when the user speaks the follow-up query to the response during the next user turn. Accordingly, these possible terms the user is likely to speak are predicted by the LLM based on the most recent response generated by the LLM and are not terms ascertainable from the conversation history nor are they included or otherwise specified in the most recent response generated by the LLM.

[0028] FIG. **1** illustrates an example system **100** for allowing a spoken conversation between a user **102** and an assistant LLM **160**. The assistant LLM **160** may be interchangeably referred to as an "LLM-powered assistant **160**", "LLM **160**", or "assistant **160**". A conversational assistant application **105** may execute on a user device **110** associated with the user **102** to enable the user **102** and the LLM-powered assistant **160** to interact with one another through spoken conversation. The conversational assistant application **105** may access various components for facilitating the spoken conversation in a natural manner between the user **102** and the LLM-powered assistant **160**. For instance, through the use of application programming interfaces (APIs) or other types of plug-ins, the conversation assistant application **105** may access a cascaded architecture that includes an automated speech recognition (ASR) system **20** including an ASR model **200**, the LLM-powered assistant **160**, and a user interface **170** including a text-to-speech (TTS) system **172**.

[0029] During a user turn of the spoken conversation between the user **102** and the LLM-powered assistant **160**, the user device **110** captures audio data **142** characterizing an utterance of a natural language query **104** spoken by the user **102** and directed toward the assistant **160** to solicit a response **165** from the assistant **160**. For instance, the query **104** may specify a particular task that the user **102** would like the assistant **160** to perform, or may specify a particular question that the user **102** would like the assistant **160** to answer and the assistant **160** may generate a response **165** that answers the question. The query **104** may similarly correspond to a request for information and the assistant **160** may generate a response **165** conveying the requested information. While the term query **104** is used, the query **104** may correspond to any natural language dialog (e.g., a greeting) directed toward the LLM-powered assistant **160** during the user's turn in the spoken conversation between the user **102** and the LLM-powered assistant **160**. The user **102** may speak the utterance of the query **104** in natural language and the ASR system **20** may perform speech recognition on the audio data **142** characterizing the utterance of the query **104** to generate a transcription **146** for the query **104** spoken by the user **102**. Thereafter, the ASR system **20** feeds the transcription **146** for the query **104** to the LLM-powered

assistant **160** to enable the LLM-powered assistant **160** to perform the task of generating a response **165** to the user's query **104**. The conversational application **105** may display a digital assistant interface **116** on a screen **117** of the user device **110** to depict a conversation history **162** of dialog turns for the conversation between the user **10** and the LLM-powered assistant **160**. The assistant LLM **160** may maintain the conversation history **162** for use as context during the voice-based conversation.

[0030] The system **100** includes the user device **110**, a remote computing system **120**, and a network **130**. The user device **110** includes data processing hardware **111** and memory hardware **112**. The user device **110** may include, or be in communication with, an audio capture device **113** (e.g., an array of one or more microphones) for converting utterances of natural language queries **104** spoken by the user **10** into corresponding audio data **102** (e.g., electrical signals or digital data). In scenarios when the user speaks a natural language query **104** captured by the microphone **113** of the user device **110**, the ASR system **20** executing on the user device **110** or the remote computing system **120** may process the corresponding audio data **102** to generate a transcription **146** of the query **104**. Here, the transcription **104** conveys the textual query **104** provided as input to the assistant LLM **160**.

[0031] The ASR system **20** includes the ASR model **210**. The ASR model **200** may implement any number and/or type(s) of past, current, or future speech recognition systems, models and/or methods including, but not limited to, an end-to-end speech recognition model, such as streaming speech recognition models having recurrent neural network-transducer (RNN-T) model architectures other frame alignment-based transducer model architectures which adhere to latency constraints associated with interactive applications, a hidden Markov model, an acoustic model, a pronunciation model, a language model, and/or a naïve Bayes classifier. In some examples, the ASR model **200** includes an audio encoder **210** (FIG. **2**) and a speech decoder **250** for decoding audio encodings **212** output by the audio encoder **210** into speech recognition results that form the transcription **146**.

[0032] Referring to FIG. **2**, an example frame alignment-based transducer model **200** includes a Recurrent Neural Network-Transducer (RNN-T) model architecture is shown. The use of the RNN-T model architecture is exemplary, and the frame alignment-based transducer model **200** may include other architectures such as transformer-transducer and conformer-transducer model architectures among others. The RNN-T model **200** provides a small computational footprint and utilizes less memory requirements than conventional ASR architectures, making the RNN-T model architecture suitable for performing speech recognition entirely on the user device **110** (e.g., no communication with a remote server is required). The RNN-T model **200** includes the audio encoder **210**, a prediction network **220**, and a joint network **230**. The audio encoder **210**, which is roughly analogous to an acoustic model (AM) in a traditional ASR system, includes a stack of self-attention layers (e.g., Conformer or Transformer layers) or a recurrent network of stacked Long Short-Term Memory (LSTM) layers. For instance, the audio encoder **210** reads a sequence of d-dimensional feature vectors (e.g., audio frames **102** (FIG. **1**) $x=(x_1, x_2, \ldots, x_T)$, where $x_t \in \mathbb{R}^d$, and produces at each output step a higher-order feature representation. This higher-order feature representation is denoted as

$$h_1^{enc}, \ldots, h_T^{enc}$$

and may be interchangeably referred to as an audio encoding **212**.

[0033] Similarly, the prediction network **220** is also an LSTM network, which, like a language model (LM), processes the sequence of non-blank symbols output by a final Softmax layer **240** so far, $y_0, \ldots, y_{ui-1}$, into a dense representation p_{u_i}. Finally, with the RNN-T model architecture, the representations produced by the encoder and prediction/decoder networks **210**, **220** are combined by the joint network **230**. The prediction network **220** may be replaced by an embedding look-up table to improve latency by outputting looked-up sparse embeddings in lieu of processing dense representations.

[0034] The joint network then predicts $P(y_i|x_{t_i}, y_0, \ldots, y_{u_{i-1}})$, which is a distribution over the next output symbol. Stated differently, the joint network **230** generates, at each output step (e.g., time step), a probability distribution over possible speech recognition hypotheses. Here, the "possible speech recognition hypotheses" correspond to a set of output labels each representing a symbol/character in a specified natural language. For example, when the natural language is English, the set of output labels may include twenty-seven (27) symbols, e.g., one label for each of the 26-letters in the English alphabet and one label designating a space. Accordingly, the joint network **230** may output a set of values indicative of the likelihood of occurrence of each of a predetermined set of output labels. This set of values can be a vector and can indicate a probability distribution over the set of output labels. In some cases, the output labels are graphemes (e.g., individual characters, and potentially punctuation and other symbols), but the set of output labels is not so limited. For example, the set of output labels can include wordpieces, phonemes, and/or entire words, in addition to or instead of graphemes. The output distribution of the joint network **230** can include a posterior probability value for each of the different output labels. Thus, if there are 100 different output labels representing different graphemes or other symbols, the output y_i of the joint network **230** can include 100 different probability values, one for each output label. The probability distribution can then be used to select and assign scores to candidate orthographic elements (e.g., graphemes, wordpieces, and/or words) in a beam search process (e.g., by the Softmax layer **240**) for determining the transcription **120**.

[0035] The Softmax layer **240** may employ any technique to select the output label/symbol with the highest probability in the distribution as the next output symbol predicted by the RNN-T model **200** at the corresponding output step. In this manner, the RNN-T model **200** does not make a conditional independence assumption, rather the prediction of each symbol is conditioned not only on the acoustics but also on the sequence of labels output so far. The RNN-T model **200** does assume an output symbol is independent of future acoustic frames **102**, which allows the RNN-T model to be employed in a streaming fashion.

[0036] In some examples, the audio encoder **210** of the RNN-T model **200** includes a stack of self-attention layers/blocks, such as conformer layers/blocks. In some examples, the number of conformer layers/blocks in the audio encoder is equal to 17 with 512-dimensional layers. The audio

encoder **210** may include 100 million parameters. Here, each conformer block includes a series of multi-headed self attention, depth wise convolution and feed-forward layers. The stack of self-attention layers/blocks may include transformer layers/blocks in other examples. The prediction network **220** may have one 512-dimensional LSTM layer. Alternatively, the prediction network **220** may include a stack of transformer or conformer blocks, or an embedding look-up table in lieu of LSTM layers. Finally, the joint network **230** may include two (2) feedforward layers with a 512-dimensional intermediate layer. The Softmax layer **240** may be composed of a unified word piece or grapheme set that is generated using all unique word pieces or graphemes in a plurality of training data sets. The Softmax layer **240** may include a 1024-dimensional layer corresponding to 1,024 wordpiece targets. The prediction network **220**, the joint network **230**, and the Softmax layer **240** may collectively form an RNN-T decoder **250** of the RNN-T model **200**. Thus, the speech decoder **250** of FIG. **1** may include the RNN-T decoder **250**

[0037] Referring back to FIG. **1**, the microphone **113** of the user device **110** may remain always-on or open during the conversation between the user **102** and the LLM-powered assistant **160** to permit the conversational application **105** to always accept speech spoken by the user **102**, thereby providing a more natural dialog between the user **102** and the assistant **160**. For instance, the user **102** may barge-in through speech directed toward the LLM-powered assistant **160** even during times when the conversational application **105** is audibly outputting, from an audio output device (e.g., speaker) **115** of the user device **110**, text-to-speech (TTS) audio **174** characterizing a response **165** generated by the LLM-powered assistant **160**.

[0038] The user device **110** may be any computing device capable of communicating with the remote computing system **120** through the network **130**. The user device **110** includes, but is not limited to, desktop computing devices and mobile computing devices, such as laptops, tablets, smart phones, smart speakers/displays, digital assistant devices, smart appliances, internet-of-things (IoT) devices, infotainment systems, vehicle infotainment systems, and wearable computing devices (e.g., headsets, smart glasses, and/or watches).

[0039] The remote computing system **120** may be a distributed system (e.g., a cloud computing environment) having scalable elastic resources. The resources include computing resources **121** (e.g., data processing hardware) and/or storage resources **122** (e.g., memory hardware). Additionally or alternatively, the remote computing system **120** may be a centralized system. The network **130** may be wired, wireless, or a combination thereof, and may include private networks and/or public networks, such as the Internet.

[0040] The components leveraged by the conversational assistant application **105** may execute on the data processing hardware **111** of the user device **110** or on the data processing hardware **121** of the remote computing system **120**. In some implementations, the components leveraged by the conversational assistant application **105** executes on both the data processing hardware **111** of the user device **110** and the data processing hardware **121** of the remote computing system **120**. For instance, one or more components of the conversational assistant application **105** may execute on the data processing hardware **111** of the user device **110** while one or more other components of the conversational assistant application **105** may execute on the remote computing system **120**.

[0041] The LLM-powered assistant **160** assistant may power the conversational assistant application **105** to function as a personal chat bot capable of having dialog conversations with the user **102** in natural language and performing tasks/actions on the user's behalf. In some examples, the LLM-powered assistant **160** includes an instance of Gemini, LaMDA, BERT, Meena, ChatGPT, Grok, or any other previously trained LLM. These previously trained LLMs have been previously trained on enormous amounts of diverse data and are capable of engaging in corresponding conversations with users in a natural and intuitive manner. However, these LLMs have a plurality of machine learning (ML) layers and hundreds of millions to hundreds of billions of ML parameters. The LLM-powered assistant **160** includes a stack of multi-head attention layers. In some examples, the stack of multi-head attention layers include Transformer layers. However, the LLM-powered assistant **160** may include other types of multi-head attention layers without departing from the scope of the present disclosure.

[0042] The conversational assistant application **105** is configured to provide, for output from the user device **110**, the response **165** generated by the LLM-powered assistant **160**. Here, the user interface **170** may audibly output, from an audio output device (e.g., acoustic speaker) **115**, text-to-speech (TTS) audio **174** that conveys the response **165** as synthesized speech. For instance, the user interface **170** may include a text-to-speech (TTS) system **172** that converts a textual representation of the response **165** into TTS audio **174** conveying the response **165** as synthesized speech. Here, the TTS system **172** may include a TTS model that converts the textual representation of the response into synthesized speech representations (e.g., Mel-frequency spectrograms) and a vocoder that convers the synthesized speech representation into time-domain audio that may be audibly output from the user device **110** as synthesized speech. Additionally or alternatively, the conversational assistant application **105** may instruct the user interface **170** to display, on the screen **117** in communication with the user device **110**, text representing the response **165**. In the example shown, the user speaks a first natural query **104**, **104***a* of "Tell a bedtime story" and the LLM-powered assistant **160** generates the response **165** of "Sure, I'd love to tell a bedtime story! Do you want to hear another spooky one?", which may be audibly output as TTS audio **174** and/or displayed in text on the screen **117**. Notably, the user interface **170** may display the conversational history **162** of queries **104** and responses **165** during the spoken conversation between the user **102** and the assistant **160**. The LLM-powered assistant **160** may maintain the conversation history **162** for use as context for generating responses **165** and may provide the conversation history **162** to the ASR model **200** for contextualizing transcriptions generated by the ASR model **200**. For instance, the conversation history **162** may include biasing terms that were recited in the previous queries **104** and responses **165** during the voice-based conversation and the conversational application may bias the ASR model **200** toward recognizing the biasing terms in user speech during the user's turn.

[0043] Continuing with the example in FIG. **1**, after the assistant LLM **160** generates the response **165** during the

assistant term, the assistant LLM **160** may implement a predictor **190** that further processes the response **165** (i.e., a textual representation of the response) to predict one or more possible terms **192** the user **102** may speak, during a next user turn in the voice-based conversation, in a follow-up query **104**, **104***b* to the response **165** generated by the LLM **160**. In some examples, parameter efficient fine-tuning (PEFT) is applied to fine-tune a small subset of parameters or individual multi-head attention layers of the assistant LLM **160** to implement the predictor **190** for predicting possible terms **192** from a LLM-generated response. In these examples, the conversational application **105** activates the small subset of parameters or individual multi-head attention layers to implement the predictor **190** responsive to the assistant LLM **160** generating the response **165** so that the predictor **190** can process the response **165** to predict the one or more possible terms **192** that the user may speak in the follow-up query **104**, **104***b* to the response **165**. Optionally, the predictor **190** may include a separate LLM **160** that is trained to predict possible terms that a user is likely to speak in a follow-up query to an LLM-generated response. In the example of FIG. **1**, the assistant LLM **160** processes the textual representation of the response **165** to determine that the response **165** includes a question directed toward the user **102** that solicits the user **102** to speak a first term (e.g. Yes) or a second term (e.g., No) in the follow-up query and determines the one or more possible terms the user may speak in the follow-up query as the first term (e.g., Yes) and the second term (e.g., No). For instance, the assistant LLM **160** provides the first term (Yes) and the second term (No) as the predicted possible terms **192** for biasing the ASR model toward recognizing Yes and No in the follow-up query **104***b* spoken by the user. Notably, the response **165** including the question solicits the user to speak the first term or the second term without explicitly specifying the first term or the second term in the response. While the example shown depicts the first term including one of "Yes" or "No" and the second term including the other one of "Yes" or "No", the first and second terms may include terms other than "Yes" and "No" without departing from the scope of the present disclosure. Moreover, while the example only depicts only two possible terms predicted by the predictor **190** of the assistant LLM **160**, the assistant LLM **160** may predict additional possible terms **192** that the user **104** may speak in the follow-on query **104***b* to the response **165** of "Sure, I'd love to tell a bedtime story! Do you want to hear another spooky one?", such as "sure", "ok", and/or "nope". Thereafter, during the user turn subsequent to the assistant turn in the voice-based conversation, the conversational application **105** receives additional audio data **142** characterizing the follow-up query **104***b* of "Yes" to the response **165** of "Sure, I'd love to tell a bedtime story! Do you want to hear another spooky one?" and the ASR model **200** biased toward recognizing the one or more possible terms **192** (e.g., "Yes" and "No") processes the additional audio data to generate a transcription **146** of the follow-on query **104***b*. The ASR model **200** may then feed the transcription **146** of the follow-on query **104***b* to assistant LLM **160** to prompt the assistant LLM **160** to commence generating a response **165** that includes a bedtime story with a spooky theme.

[0044] In addition to, or in lieu, of predicting possible terms **192** the user **102** may speak in the follow-on query **104***b*, the predictor **190** of the assistant LLM **160** may process the textual representation of the response **165** to determine an expectation of a type of query **104** or a number range of words the user **102** will speak during the next user turn after the response **165** is output to the user **104**. For instance, the predictor **190** may process the textual representation of the response **165** of "Sure, I'd love to tell a bedtime story! Do you want to hear another spooky one?" and determine the expectation of a short follow-up query (e.g., less than five words) based on presence of the "?" in the response **165**. In other example, predictor **190** may determine the expectation of the follow-up query including dictated speech. Similarly, the predictor **190** could process a textual representation of another response **165** of "To practice your Spanish, please do your best to speak 'Summer is my favorite season' in Spanish" and determine the expectation that the follow-up query will include speech spoken in the language Spanish. Here, the ASR model **200** is a multilingual ASR model and the LLM **160** could provide a language identifier indicating that the ASR model **200** should bias toward recognizing Spanish.

[0045] In another example, FIGS. **3**A-**3**C provide schematic views of user and assistant turns **300***a-c* during another voice-based conversation between the user **102** and the assistant LLM **160**. Referring to FIG. **3**A, during a first user turn **300***a* in the voice-based conversation, the user **10** speaks a first query **104***a* of "Tell a bedtime story" directed toward the assistant LLM **160**. The ASR model **200** processes audio data **142** characterizing the first query **104***a* to generate a transcription **146** of the first query **104***a*. Thereafter, the ASR model **200** feeds the transcription **146** of the first query **104***a* to the assistant LLM **160** to initiate a next assistant turn **300***b*.

[0046] Referring to FIG. **3**B, during the assistant turn **300***b* in the voice-based conversation, the assistant LLM **160** processes the transcription **146** of the first query **104***a* to generate a response **165** of "Sure! Is there a particular type of bedtime story you'd like to hear?". While not shown, the response **165** may be provided to the user interface **170** for audibly outputting TTS audio **172** that conveys the response **165** as synthesized speech and/or graphically displaying the response **165** as text. After generating the response **165**, the predictor **190** processes a textual representation of the response **165** to determine that the response **165** includes a question that solicits the user to speak an answer to the question and identify a list of likely answers to the question. In the example shown, based on the response **165** requesting the user **102** to input a particular type of story, the predictor **192** generates the list of likely answers to include different types of bedtime stories such as spooky, classic, fairy tale, funny, calming, etc. . . . Accordingly, the assistant LLM **160** determines the one or more possible terms **192** the user **102** may speak in a follow-up query as the list of likely answers to the question and biases the ASR model **200** to bias the ASR model **200** toward recognizing the possible terms **192** that includes the list of likely answers pertaining to different types of bedtime stories.

[0047] Referring to FIG. **3**C, during a second user turn **300***c* subsequent to the assistant turn **300***b* in the voice-based conversation, the user **102** speaks a second query **104***b* of "Please tell a spooky one" directed toward the assistant LLM indicating that the user **102** would like the assistant LLM **160** to tell a spooky bedtime story. Here, the ASR model **200** biased toward recognizing the possible terms **182** that includes the list of likely answers (e.g., spooky, classic, fairy tale, funny, calming) processes audio data character-

izing the follow-up query **104***b* to generate a transcription **146** of the second query **104***b*. Thereafter, the ASR model **200** feeds the transcription **146** of the second query **104***b* to the assistant LLM **160** to prompt the assistant LLM **160** to commence generating the bedtime story with the spooky them during a subsequent assistant turn in the voice-based conversation.

[0048] FIG. **4** includes a flowchart of an example arrangement of operations for a computer-implemented **400** of biasing an ASR model **200** during a voice-based conversation between a user **102** and an assistant LLM **160**. The method **400** may execute on data processing hardware **510** (FIG. **5**) using instructions stored on memory hardware **520** (FIG. **5**) that may reside on the user device **110** and/or the remote system **120** of FIG. **1** each corresponding to a computing device **500** (FIG. **5**).

[0049] At operation **402**, during an assistant turn in the voice-based conversation between the user and the assistant LLM, the method **400** includes receiving a response **165** generated by the assistant LLM **160** that is directed toward the user. At operation **404**, based on the response **165** generated by the assistant LLM **160** that is directed toward the user, the method **400** includes predicting, by the assistant LLM **160**, one or more possible terms **192** the user may speak in a follow-up query **104** to the response **165** generated by the assistant LLM **165** during a user turn subsequent to the assistant turn in the voice-based conversation.

[0050] At operation **406**, the method **400** includes biasing the ASR model **200** toward recognizing the one or more possible terms **192** predicted by the assistant LLM **160**. At operation **408**, during the user turn subsequent to the assistant turn in the voice-based conversation, the method **400** includes receiving audio data **142** characterizing the follow-up query **104** to the response **165** generated by the assistant LLM **160**. At operation **410**, the method **400** includes processing, using the ASR model **200** biased toward recognizing the one or more possible terms **192** predicted by the assistant LLM **160**, the audio data **142** to generate a transcription **146** of the follow-up query **104**.

[0051] FIG. **5** is a schematic view of an example computing device **500** that may be used to implement the systems and methods described in this document. The computing device **500** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or claimed in this document.

[0052] The computing device **500** includes a processor **510**, memory **520**, a storage device **530**, a high-speed interface/controller **540** connecting to the memory **520** and high-speed expansion ports **550**, and a low speed interface/controller **560** connecting to a low speed bus **570** and a storage device **530**. Each of the components **510**, **520**, **530**, **540**, **550**, and **560**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **510** can process instructions for execution within the computing device **500**, including instructions stored in the memory **520** or on the storage device **530** to display graphical information for a graphical user interface (GUI) on an external input/output device, such as display **580** coupled to high speed interface **540**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **500** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

[0053] The memory **520** stores information non-transitorily within the computing device **500**. The memory **520** may be a computer-readable medium, a volatile memory unit(s), or non-volatile memory unit(s). The non-transitory memory **520** may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by the computing device **500**. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

[0054] The storage device **530** is capable of providing mass storage for the computing device **500**. In some implementations, the storage device **530** is a computer-readable medium. In various different implementations, the storage device **530** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. In additional implementations, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **520**, the storage device **530**, or memory on processor **510**.

[0055] The high speed controller **540** manages bandwidth-intensive operations for the computing device **500**, while the low speed controller **560** manages lower bandwidth-intensive operations. Such allocation of duties is exemplary only. In some implementations, the high-speed controller **540** is coupled to the memory **520**, the display **580** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **550**, which may accept various expansion cards (not shown). In some implementations, the low-speed controller **560** is coupled to the storage device **530** and a low-speed expansion port **590**. The low-speed expansion port **590**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

[0056] The computing device **500** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **500***a* or multiple times in a group of such servers **500***a*, as a laptop computer **500***b*, or as part of a rack server system **500***c*.

[0057] Various implementations of the systems and techniques described herein can be realized in digital electronic

and/or optical circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

[0058] These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms "machine-readable medium" and "computer-readable medium" refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term "machine-readable signal" refers to any signal used to provide machine instructions and/or data to a programmable processor.

[0059] The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

[0060] To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

[0061] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

What is claimed is:

1. A computer-implemented method executing on data processing hardware that causes the data processing hardware to perform operations comprising:

during an assistant turn in a voice-based conversation between a user and an assistant large language model (LLM), receiving a response generated by the assistant LLM that is directed toward the user;

based on the response generated by the assistant LLM that is directed toward the user, predicting, by the assistant LLM, one or more possible terms the user may speak in a follow-up query to the response generated by the assistant LLM during a user turn subsequent to the assistant turn in the voice-based conversation;

biasing an automated speech recognition (ASR) model toward recognizing the one or more possible terms predicted by the assistant LLM;

during the user turn subsequent to the assistant turn in the voice-based conversation, receiving audio data characterizing the follow-up query to the response generated by the assistant LLM; and

processing, using the ASR model biased toward recognizing the one or more possible terms predicted by the assistant LLM, the audio data to generate a transcription of the follow-up query.

2. The computer-implemented method of claim **1**, wherein predicting the one or more possible terms the user may speak in the follow-up query comprises:

processing a textual representation of the response to:

determine that the response directed toward the user comprises a question that solicits the user to speak an answer to the question; and

identify a list of likely answers to the question; and

determining the one or more possible terms the user may speak in the follow-up query as the list of likely answers to the question.

3. The computer-implemented method of claim **1**, wherein predicting the one or more possible terms the user may speak in the follow-up query comprises:

processing a textual representation of the response to determine the response comprises a question directed toward the user that solicits the user to speak a first term or a second term in the follow-up query; and

determining the one or more possible terms the user may speak in the follow-up query comprises the first term and the second term.

4. The computer-implemented method of claim **3**, wherein:

the first term comprises one of Yes or No; and

the second term comprises the other one of Yes or No.

5. The computer-implemented method of claim **3**, wherein the response comprising the question directed toward the user solicits the user to speak the first term or the second term without specifying the first term and the second term in the response.

6. The computer-implemented method of claim **1**, wherein the operations further comprise:

obtaining, from the assistant LLM, a conversation history of the voice-based conversation comprising all previous queries input by the user and corresponding responses returned by the assistant LLM during the voice-based conversation,

wherein biasing the ASR model toward recognizing the one or more possible terms predicted by the assistant LLM further comprises biasing the ASR model toward recognizing biasing terms related to the conversation history of the voice-based conversation.

7. The computer-implemented method of claim **6**, wherein the one or more possible terms predicted by the assistant LLM are different from the biasing terms related to the conversation history.

8. The computer-implemented method of claim **1**, wherein the operations further comprise, after generating the transcription of the follow-up query, processing, by the assistant LLM, the transcription of the follow-up query to generate another response directed toward the user that is responsive to the follow-up query.

9. The computer-implemented method of claim **1**, wherein the operations further comprise:

processing, by a text-to-speech (TTS) system, a textual representation of the response generated by assistant LLM to generate TTS audio characterizing a synthesized speech representation of the response; and

providing, for audible output from a user device associated with the user, the TTS audio characterizing the synthesized speech representation of the response.

10. The computer-implemented method of claim **1**, wherein:

the ASR model comprises:

an acoustic encoder; and

a speech decoder;

the assistant LLM comprises a plurality of pre-trained multi-head attention layers; and

the ASR model and the plurality of pre-trained multi-head attention layers are trained separately.

11. A system comprising:

data processing hardware; and

memory hardware in communication with the data processing hardware, the memory hardware storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising:

during an assistant turn in a voice-based conversation between a user and an assistant large language model (LLM), receiving a response generated by the assistant LLM that is directed toward the user;

based on the response generated by the assistant LLM that is directed toward the user, predicting, by the assistant LLM, one or more possible terms the user may speak in a follow-up query to the response generated by the assistant LLM during a user turn subsequent to the assistant turn in the voice-based conversation;

biasing an automated speech recognition (ASR) model toward recognizing the one or more possible terms predicted by the assistant LLM;

during the user turn subsequent to the assistant turn in the voice-based conversation, receiving audio data characterizing the follow-up query to the response generated by the assistant LLM; and

processing, using the ASR model biased toward recognizing the one or more possible terms predicted by the assistant LLM, the audio data to generate a transcription of the follow-up query.

12. The system of claim **11**, wherein predicting the one or more possible terms the user may speak in the follow-up query comprises:

processing a textual representation of the response to:

determine that the response directed toward the user comprises a question that solicits the user to speak an answer to the question; and

identify a list of likely answers to the question; and

determining the one or more possible terms the user may speak in the follow-up query as the list of likely answers to the question.

13. The system of claim **11**, wherein predicting the one or more possible terms the user may speak in the follow-up query comprises:

processing a textual representation of the response to determine the response comprises a question directed toward the user that solicits the user to speak a first term or a second term in the follow-up query; and

determining the one or more possible terms the user may speak in the follow-up query comprises the first term and the second term.

14. The system of claim **13**, wherein:

the first term comprises one of Yes or No; and

the second term comprises the other one of Yes or No.

15. The system of claim **13**, wherein the response comprising the question directed toward the user solicits the user to speak the first term or the second term without specifying the first term and the second term in the response.

16. The system of claim **11**, wherein the operations further comprise:

obtaining, from the assistant LLM, a conversation history of the voice-based conversation comprising all previous queries input by the user and corresponding responses returned by the assistant LLM during the voice-based conversation,

wherein biasing the ASR model toward recognizing the one or more possible terms predicted by the assistant LLM further comprises biasing the ASR model toward recognizing biasing terms related to the conversation history of the voice-based conversation.

17. The system of claim **16**, wherein the one or more possible terms predicted by the assistant LLM are different from the biasing terms related to the conversation history.

18. The system of claim **11**, wherein the operations further comprise, after generating the transcription of the follow-up query, processing, by the assistant LLM, the transcription of the follow-up query to generate another response directed toward the user that is responsive to the follow-up query.

19. The system of claim **11**, wherein the operations further comprise:

processing, by a text-to-speech (TTS) system, a textual representation of the response generated by assistant LLM to generate TTS audio characterizing a synthesized speech representation of the response; and

providing, for audible output from a user device associated with the user, the TTS audio characterizing the synthesized speech representation of the response.

20. The system of claim **11**, wherein:

the ASR model comprises:

an acoustic encoder; and

a speech decoder;

the assistant LLM comprises a plurality of pre-trained multi-head attention layers; and

the ASR model and the plurality of pre-trained multi-head attention layers are trained separately.

* * * * *